

# **Advanced Architecture and Implementation Lifecycle for Medical AI Systems: Integrating Mixture-of-Experts, Agent Swarms, and Conditional Memory**

The transition from static, rule-based medical symptom checkers to autonomous, generative artificial intelligence represents a paradigm shift in healthcare diagnostics and clinical logistics. Historical systems relying on decision trees or basic keyword associations have consistently demonstrated severe limitations, notably in diagnostic accuracy, often failing to exceed a thirty-five percent correct primary diagnosis rate.<sup>1</sup> While large language models offer probabilistic flexibility and have achieved expert-level scores on medical licensing examinations, conventional monolithic architectures struggle in production healthcare environments. They are prone to hallucination, suffer from restricted context windows, and lack the targeted deterministic recall required for stringent clinical safety.<sup>1</sup>

The preliminary architecture of the proposed medical AI application, referred to as Sushrut AI, establishes a foundational framework by utilizing a hub-spoke topology. This design features a Central Generalist Agent that acts as a primary triage coordinator, routing tasks to thirteen specialized domain sub-agents trained on specific clinical data.<sup>1</sup> While logical for initial development, advancing this framework to a production-grade clinical ecosystem requires integrating sophisticated scaling mechanisms. The objective is to preserve the central controlling agent while fusing it with decentralized agent swarm logic, supported by advanced foundation models incorporating Mixture-of-Experts, hybrid attention mechanisms, and conditional memory. The following analysis details the scope of architectural improvements, proposes a final integrated architecture, and provides an exhaustive, step-by-step implementation guideline from initial innovation to final clinical deployment.

## **Analysis of Preliminary Design and Scope of Improvements**

The baseline architecture relies on a centralized manager-worker hierarchy where a central agent orchestrates sub-agents utilizing standard Retrieval-Augmented Generation to mitigate hallucinations.<sup>1</sup> While this addresses basic clinical routing, scaling this system to support millions of concurrent users, process lifetime electronic health records, and operate within the bounds of stringent clinical regulatory frameworks necessitates the resolution of several critical bottlenecks.

In a strictly hierarchical multi-agent system, the central supervisor frequently becomes a single point of failure and a severe computational bottleneck.<sup>3</sup> If a patient presents with overlapping, multi-systemic symptoms requiring multi-disciplinary collaboration, the central router must synthesize all intermediate outputs sequentially. This forces the central agent's context window to overflow, leading to reasoning degradation and semantic drift.<sup>3</sup> Furthermore, relying entirely on a central node prevents specialized agents from iterating on each other's findings dynamically. The improvement required here is the integration of a hierarchical swarm pattern, which retains the central controller for final synthesis and user interaction but allows the sub-agents to operate as a decentralized swarm for parallel problem-solving.<sup>5</sup>

The preliminary architecture's reliance on standard vector databases for Retrieval-Augmented Generation introduces latency and retrieval noise. In clinical settings, fetching fixed knowledge such as pharmacological interactions, exact drug dosages, or genetic sequences via semantic similarity search is computationally inefficient and imprecise compared to deterministic lookups.<sup>1</sup> Semantic search can retrieve tangentially related medical texts that confuse the generation process, leading to conditionally correct but medically inappropriate outputs.<sup>8</sup> The integration of the Engram conditional memory module is required to replace or augment probabilistic retrieval with constant-time, deterministic factual lookups.<sup>1</sup>

Finally, the underlying foundation model powering the agents must be upgraded. Utilizing a dense large language model for thirteen distinct clinical agents is computationally prohibitive. The architecture requires a shift to a sparse Mixture-of-Experts framework, which decouples the model's total knowledge capacity from its active computational cost.<sup>1</sup> By integrating the DeepSeek-V4 architecture, the system can support the required multi-agent reasoning while simultaneously managing a one-million-token context window, allowing the agents to analyze a patient's entire lifetime medical history instantaneously.<sup>1</sup>

## **Final Suggested Architecture: Hierarchical Swarm with Centralized Orchestration**

To satisfy the requirement of maintaining a central controlling agent while maximizing the efficiency of the total system, the final architecture must adopt a Hierarchical Swarm pattern. This hybrid paradigm marries the structured accountability of centralized orchestration with the parallel, emergent reasoning capabilities of swarm intelligence.<sup>6</sup>

The Central Generalist Agent functions as the executive orchestrator and the exclusive interface for the patient or clinician. Upon receiving a complex medical query or unstructured electronic health record data, the Central Agent performs initial natural language processing and named entity recognition to estimate the depth of the case.<sup>1</sup> Instead of merely routing the task to a single sub-agent, the Central Agent dynamically decomposes the clinical problem into parallel subtasks and dispatches them to a customized swarm of the thirteen specialized sub-agents.<sup>1</sup>

Once dispatched, the central controller temporarily steps back, allowing the specialized sub-agents to collaborate using a decentralized Blackboard architecture.<sup>12</sup> In this configuration, the sub-agents share a common memory space. For instance, the Pathology Agent interprets laboratory results and posts the structured findings to the blackboard. The Endocrinology Agent reads these results, identifies a metabolic anomaly, and posts an updated differential diagnosis. Concurrently, the Pharmacology Agent evaluates the proposed treatments against the patient's existing medication list for contraindications.<sup>13</sup> This peer-to-peer collaboration enables the swarm to explore the diagnostic solution space iteratively, critiquing and refining each other's outputs without waiting for the central agent to process every intermediate step.<sup>3</sup>

To ensure the swarm eventually reaches a consensus, the Central Agent monitors the blackboard for specific exit conditions, such as a stabilized differential diagnosis or a time limit.<sup>11</sup> Once the exit condition is met, the Central Agent synthesizes the swarm's collective findings, formats the output according to clinical guidelines, and delivers the final diagnostic report or care logistics plan to the user. This architecture prevents the central orchestrator from becoming a computational bottleneck while preserving strict oversight and a unified user experience.<sup>10</sup>

## **Foundation Model Layer: Mixture-of-Experts**

Powering this complex multi-agent orchestration requires a foundation model of immense capacity and efficiency. The integration of the DeepSeek-V4 Mixture-of-Experts architecture fulfills this requirement.<sup>1</sup> MoE scales model capacity by sparsely activating parameters to process dynamic logic, serving as a paradigm for conditional computation.<sup>1</sup> Models such as DeepSeek-V4-Pro utilize 1.6 trillion total parameters while activating only 49 billion parameters per token.<sup>1</sup> This separation of total capacity from active computation allows the medical AI to store encyclopedic medical knowledge across hundreds of highly specialized routed experts without incurring prohibitive computational costs during inference.<sup>1</sup>

To handle the immense data load of multi-turn medical diagnostics and lifetime patient histories, the architecture utilizes a Hybrid Attention Mechanism designed to support a one-million-token context window.<sup>1</sup> Traditional attention mechanisms scale quadratically, making comprehensive electronic health record analysis impossible. The hybrid approach resolves this by combining Compressed Sparse Attention and Heavily Compressed Attention.<sup>1</sup>

Compressed Sparse Attention compresses Key-Value caches by a specific factor. Given input hidden states, the mechanism computes two series of Key-Value entries and their corresponding compression weights. A defined number of entries are compressed into a single entry using softmax-normalized weights and learnable positional biases. A lightning indexer then evaluates the queries and selects the optimal compressed entries for Multi-Query Attention.<sup>1</sup> Heavily Compressed Attention applies an even larger compression rate to consolidate Key-Value caches, operating without sparse selection to maintain dense attention over heavily summarized historical states.<sup>1</sup> By interleaving these mechanisms, the foundation model requires only twenty-seven percent of the single-token inference compute compared to

standard models, enabling the central agent and the swarm to retain massive clinical contexts actively.<sup>1</sup>

To guarantee signal stability across these extreme context lengths, Manifold-Constrained Hyper-Connections are integrated into the architecture. Traditional residual connections frequently cause numerical instability in deep networks. The constrained connections restrict the residual mapping matrix to the Birkhoff polytope, which is the mathematical manifold of doubly stochastic matrices. This ensures that the spectral norm of the mapping matrix remains bounded, preventing exploding activations when the swarm analyzes massive, million-token patient dossiers.<sup>1</sup>

## **Conditional Memory Layer: Engram Integration**

While the Mixture-of-Experts architecture handles conditional computation and complex diagnostic reasoning, medical systems require absolute, deterministic precision for factual data. Transformers inherently lack a native primitive for knowledge lookup, forcing the model to simulate retrieval through computationally expensive neural pathways in early layers.<sup>1</sup> To resolve this, the architecture integrates the Engram conditional memory module, which provides constant-time lookup for static patterns by modernizing N-gram embeddings.<sup>1</sup> This acts as a structural complement to the neural reasoning layers, significantly outperforming standard probabilistic retrieval architectures for factual medical data.

Engram operates through deterministic hashing to separate static pattern storage from dynamic computation. A vocabulary projection layer compresses raw token identifiers into canonical identifiers via normalization techniques, significantly reducing vocabulary fragmentation for complex medical terminology.<sup>1</sup> For a given local context, multi-head hashing utilizes distinct hash heads to map the context to an index within a massive embedding table.<sup>1</sup> The retrieved static medical embeddings are subsequently fused dynamically into the Transformer backbone utilizing context-aware gating. The current hidden state acts as a query to modulate the retrieved memory, ensuring that the fetched medical facts align semantically with the ongoing diagnostic reasoning.<sup>1</sup>

Because Engram relies on deterministic addressing rather than dynamic routing, massive medical embedding tables containing billions of parameters derived from medical textbooks and pharmacological databases can be offloaded to host central processing unit memory.<sup>1</sup> This allows the AI swarm to fetch exact medical definitions, ICD-10 codes, and drug interaction data with negligible inference overhead, completely bypassing graphics processing unit memory constraints.<sup>1</sup> Mechanistic analyses indicate that delegating these local factual dependencies to deterministic lookups relieves the early layers of the network from static reconstruction, effectively deepening the network and freeing attention capacity for the global clinical context.<sup>1</sup>

| Architectural Component | Underlying Technology            | Clinical Application within the Framework                                                                               |
|-------------------------|----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Central Orchestrator    | Hierarchical Swarm Dispatcher    | Evaluates patient input, decomposes clinical tasks, and manages swarm exit conditions and final synthesis.              |
| Swarm Intelligence      | Blackboard Pattern Collaboration | Enables the thirteen sub-agents to debate and refine differential diagnoses in parallel without bottlenecks.            |
| Foundation Model        | DeepSeek-V4 Mixture-of-Experts   | Provides 1.6 trillion parameters of medical capacity while maintaining low inference latency via sparse activation.     |
| Context Management      | Hybrid Attention (CSA/HCA)       | Supports one-million-token context windows for analyzing lifetime longitudinal patient histories.                       |
| Factual Retrieval       | Engram Conditional Memory        | Replaces noisy semantic search with deterministic, constant-time lookups for exact pharmacological and anatomical data. |
| Data Integration        | Model Context Protocol (MCP)     | Enforces strict, auditable security boundaries for agents accessing external electronic                                 |

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|  |  | health records. |
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# Step-by-Step Implementation Guideline: Innovation to Product Level

Developing and deploying an architecture of this complexity within the highly regulated healthcare sector requires an exhaustive, multi-phase lifecycle. The following guideline details the progression from initial data curation and base model training to clinical validation, continuous reinforcement learning, and final product deployment.

## Phase One: Base Model Initialization and Continued Pre-Training

The implementation commences with the initialization of the foundation model and the execution of Continued Pre-Training to inject encyclopedic medical knowledge into the parameters.<sup>1</sup> General-purpose models lack the nuanced semantic understanding required for precision medicine. The training corpus must be aggressively curated from high-fidelity sources to establish the fundamental knowledge of an expert generalist.

The primary datasets for this phase include medical textbooks, encyclopedias, and literature corpora such as MedQA and Medical Meadow WikiDoc, which provide comprehensive anatomical, physiological, and pathological grounding.<sup>1</sup> To ensure the model understands pharmacological constraints, databases such as DailyMed and OpenFDA are ingested to provide granular details on drug labels, dosages, and contraindications.<sup>1</sup> Real-world clinical reasoning logic is injected utilizing large-scale repositories of anonymized clinical notes, such as the MIMIC-IV database, which contains hundreds of thousands of discharge summaries.<sup>1</sup>

During this phase, the Engram conditional memory module is trained alongside the neural weights. The tokenizer is optimized via the Engram vocabulary projection layer to compress multi-word medical entities into canonical identifiers, increasing the overall semantic density of the medical corpus.<sup>1</sup>

Training trillion-parameter Mixture-of-Experts models on complex medical data frequently introduces numerical instability and loss spikes. To accelerate convergence and maintain gradient stability, the training pipeline utilizes the Muon optimizer.<sup>1</sup> For logically independent weight matrices, the Muon optimizer calculates gradients, accumulates a momentum buffer, and performs orthogonalization via hybrid Newton-Schulz iterations. This process strictly regulates the update matrices to prevent divergence.<sup>1</sup> Furthermore, to prevent loss spikes caused by expert routing imbalances, Anticipatory Routing is implemented. The routing indices for the current step are pre-computed using historical parameters, decoupling the synchronous updates of the backbone and the routing network. SwiGLU Clamping is also applied to confine the linear component of the activation function, explicitly eliminating outliers

during medical data ingestion.<sup>1</sup>

## **Phase Two: Agentic Post-Training and Swarm Synthesis**

Following the establishment of the base foundation model, the system advances to a two-stage post-training pipeline designed to cultivate the thirteen specialized sub-agents and consolidate their capabilities for swarm collaboration.

The first stage involves Specialist Training and Supervised Fine-Tuning for each distinct clinical domain.<sup>1</sup> Each of the thirteen agents, representing domains such as Oncology, Pulmonology, and Gastroenterology, is independently fine-tuned on highly specific data splits derived from the HPAI-BSC medical specialties dataset.<sup>1</sup> This training utilizes Instruction-Response pairs to teach the models specific behavioral reasoning. For instance, the ChatDoctor dataset is utilized to instill clinical dialogue empathy and bedside manner, while PubMedQA provides a foundation for rigorous, evidence-based scientific reasoning.<sup>1</sup> During this stage, the agents are trained to utilize specific demarcation tags to outline their internal diagnostic reasoning pathways explicitly prior to outputting a final clinical decision, ensuring that their logic is transparent and auditable.<sup>1</sup>

To prepare these individual agents to operate efficiently within a swarm architecture and utilize external medical tools, the pipeline employs Large-Scale Agentic Task Synthesis.<sup>1</sup> A specialized environment-synthesis agent constructs thousands of virtual patient profiles and simulated electronic health record databases. Task-specific tools are synthesized, representing the precise application programming interfaces the agents will use in production. The pipeline systematically generates tens of thousands of complex clinical prompts, forcing the agents to iteratively retrieve data, analyze simulated laboratory results, and adjust their differential diagnoses in a closed-loop sandbox.<sup>1</sup>

Once the thirteen specialized agents demonstrate proficiency in their respective domains and tool usage, their capabilities are merged into a unified parameter space via multi-teacher On-Policy Distillation.<sup>1</sup> The unified foundation model acts as a student, minimizing the reverse Kullback-Leibler divergence between its predictions and the logits generated by the specialized teachers. Full-vocabulary logit distillation is utilized to maintain gradient stability and ensure faithful transfer of the teachers' nuanced clinical knowledge. To manage the massive graphics processing unit memory footprint of thirteen simultaneous teacher models, the infrastructure employs efficient teacher scheduling, caching only the last-layer hidden states of the teachers and reconstructing the required logits dynamically during training.<sup>1</sup>

## **Phase Three: Continuous Optimization via Agent Lightning**

A defining characteristic of an advanced medical AI system is its ability to continuously learn and adapt its clinical reasoning based on real-world interactions and shifting medical guidelines. Traditional Reinforcement Learning from Human Feedback requires extensive, resource-intensive retraining cycles and tightly couples the training algorithm to the agent

logic, making it unsuitable for a live production swarm.<sup>18</sup>

To achieve continuous, non-intrusive optimization, the architecture implements Microsoft's Agent Lightning framework.<sup>18</sup> Agent Lightning utilizes a Training-Agent Disaggregation architecture, acting as robust middleware between the reinforcement learning algorithms and the executing agent environment.<sup>20</sup> This architecture is built upon three core components that function in a coordinated loop. The LightningStore serves as a centralized hub and message queue, maintaining tasks, resources, and execution traces. The Runner acts as the execution worker, pulling clinical rollout tasks, operating the agent swarm, and streaming the results back to the store. The Algorithm functions as the learning core, enqueueing new scenarios, evaluating the collected traces, and computing updates for the model.<sup>22</sup>

As the central agent and the swarm interact with simulated or anonymized real-world clinical environments, Agent Lightning uses a sidecar-based monitoring approach to capture telemetry non-intrusively without altering the underlying agent codebase.<sup>18</sup> It meticulously records execution traces, agent-to-agent blackboard communications, external tool invocations, and any clinical logic errors.

These comprehensive traces are converted into Markov Decision Process transitions. Rather than relying on a traditional scalar reward model, which struggles with the nuanced complexities of hard-to-verify medical diagnoses, the system employs a Generative Reward Model.<sup>1</sup> The actor network evaluates its own clinical trajectories against rigorous, predefined medical rubrics. The Group Relative Policy Optimization algorithm analyzes these evaluations and iteratively updates the system's prompt templates, tool-call logic, and internal model weights.<sup>1</sup> This continuous reinforcement learning loop allows the swarm to dynamically refine its multi-agent coordination, heavily reduce the occurrence of diagnostic hallucination, and optimize complex clinical decision-making over time.<sup>18</sup>

| Post-Training Methodology | Framework Utilized                  | Objective within the Clinical System                                                      |
|---------------------------|-------------------------------------|-------------------------------------------------------------------------------------------|
| Supervised Fine-Tuning    | Domain-Specific Datasets (HPAI-BSC) | Instills specialized medical knowledge and empathy into the thirteen distinct sub-agents. |
| Agentic Synthesis         | Simulated Sandbox Environments      | Teaches the agents to utilize external clinical tools and navigate multi-step diagnostic  |

|                            |                                 |                                                                                                     |
|----------------------------|---------------------------------|-----------------------------------------------------------------------------------------------------|
|                            |                                 | workflows.                                                                                          |
| <b>Model Consolidation</b> | On-Policy Distillation (OPD)    | Merges the specialized capabilities of the agents into a unified, highly efficient parameter space. |
| <b>Continuous Learning</b> | Microsoft Agent Lightning       | Enables non-intrusive reinforcement learning to optimize swarm coordination without code rewrites.  |
| <b>Reward Optimization</b> | Generative Reward Models & GRPO | Replaces basic scalar rewards with nuanced, rubric-based clinical evaluations for complex cases.    |

## Phase Four: Clinical Validation and Regulatory Compliance

Deploying autonomous, non-deterministic agent swarms in healthcare requires strict adherence to international regulatory frameworks. Unlike simple static diagnostic algorithms, generative multi-agent systems exhibit emergent behavior and probabilistic outputs, classifying them as Unconfined Non-Deterministic Clinical Software.<sup>25</sup> The system must comply with the United States Food and Drug Administration guidelines for Software as a Medical Device and internationally recognized quality standards.<sup>26</sup>

The software development lifecycle must be strictly governed by the IEC 62304 standard, which dictates comprehensive design control, risk management integration, and structured lifecycle documentation for medical software.<sup>28</sup> These software procedures must be fully enveloped within the broader ISO 13485 Quality Management System, ensuring that all continuous integration pipelines, reinforcement learning weight updates, and testing protocols are fully traceable and auditable during regulatory inspections.<sup>28</sup> Concurrently, continuous risk management processes must be aligned with the ISO 14971 standard.<sup>29</sup>

Because the medical swarm continuously learns and adapts its behavior via the Agent Lightning framework, it requires a robust regulatory strategy under the Food and Drug

Administration's Total Product Lifecycle approach.<sup>32</sup> To avoid the prohibitive burden of submitting a new 510(k) clearance application for every minor, automated weight update, the deployment strategy must utilize a Predetermined Change Control Plan.<sup>33</sup> Authorized through the Q-Submission process, the predetermined plan provides the regulatory body with a preauthorized playbook for future algorithmic updates.<sup>34</sup> This plan requires three interlocking components: a detailed description of the modifications defining the exact scope of the automated updates; a modification protocol detailing exactly how the Agent Lightning framework collects data, trains, and validates changes automatically; and a comprehensive impact assessment evaluating the cumulative risks, benefits, and bias mitigations associated with continuous learning.<sup>34</sup>

To ensure absolute clinical safety at runtime, the outputs of the swarm are bounded by the Healthcare AI Agents Regulatory Framework.<sup>36</sup> This framework introduces multi-stage deterministic middleware to enforce rigorous safety boundaries. Before any diagnostic output or logistical care plan is passed from the central agent to a user or physician, it undergoes dual risk evaluation. The Clinical Safety Risk Assessment evaluates the medical risk of the proposed action based on established principles, while the Hallucination Risk Assessment measures factual reliability.<sup>8</sup> The hallucination assessment calculates the Shannon entropy of the generated log-probabilities to detect semantic drift and uncertainty; high entropy indicates the model is guessing between low-probability tokens, a clear sign of factual failure.<sup>2</sup> Responses are only released to the user if both safety and hallucination scores fall below strict clinical acceptability thresholds.<sup>8</sup> The deterministic middleware ensures a zero percent success rate for adversarial prompt-injection and unauthorized medical tool use.<sup>36</sup>

## **Phase Five: Production Deployment and Care Logistics Integration**

The final phase of the implementation lifecycle transitions the validated system into live production, requiring specialized inference infrastructure and the integration of logistical care protocols. Processing continuous, one-million-token contexts for thousands of concurrent patients requires aggressive infrastructural optimization to prevent latency degradation.<sup>1</sup>

The inference framework isolates the Key-Value cache into a heterogeneous structure, managing the Compressed Sparse Attention and Heavily Compressed Attention states separately to maximize hardware utilization.<sup>1</sup> To minimize the computational cost of multi-turn medical interactions that span days or weeks, an On-Disk Key-Value Cache Storage mechanism is deployed. The compressed cache entries for a patient's historical records are serialized to high-speed solid-state drives.<sup>1</sup> When a patient or clinician resumes a session, the system asynchronously prefetches the shared-prefix cache directly from the disk, overlapping input-output operations with active computation to provide near-instantaneous response times.<sup>1</sup> For the uncompressed Sliding Window Attention caches, periodic checkpointing allows the system to balance storage overhead with minimal recomputation redundancy.<sup>1</sup>

For the agent swarm to interface safely with external healthcare infrastructure—such as electronic health records, laboratory information systems, and imaging databases—it relies

exclusively on the Model Context Protocol.<sup>38</sup> The protocol provides a universal, secure, and standardized two-way connection between the AI agents and external data sources.<sup>39</sup> In a highly regulated medical environment, the protocol enforces strict security boundaries and tenant-aware data segregation, ensuring that agents only extract data they are explicitly authorized to view.<sup>41</sup> By translating raw database queries into structured, verified responses, the protocol prevents agents from hallucinating patient data and provides an auditable cryptographic trail of every data request, fulfilling the logging requirements of healthcare privacy laws.<sup>41</sup>

Advanced medical AI systems must transcend isolated digital diagnostics to execute tangible, physical care delivery.<sup>1</sup> The interaction layer of the architecture facilitates end-to-end logistics through dedicated utility agents that operate alongside the clinical diagnostic swarm. A Concierge Agent manages complex medical tourism requirements, seamlessly coordinating cross-border logistics, hospital admissions, and transportation tracking for patients seeking specialized care.<sup>1</sup> A Pharmacy Agent verifies generated prescriptions against the Engram-backed pharmacology database for safety, interfaces with clinical inventory systems via the Model Context Protocol, and orchestrates final medication delivery.<sup>1</sup> Furthermore, an Emergency Alert Agent continuously monitors real-time patient telemetry through application programming interface feeds from wearable devices; it can autonomously dispatch emergency services if its critical threshold algorithms detect acute physiological deterioration.<sup>1</sup> By executing these operations via standardized, secure protocols, the logistical agents interact safely with third-party vendors and hospital networks, ensuring a seamless continuity of care from initial symptom triage to final treatment execution.

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